

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

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Under a rigid-strain upper-bound protocol with fixed fractional coordinates, reported periodic $|\mathcal{S}|$ magnitudes serve as upper bounds—not predictions—of equilibrium total-energy cross terms until sign-qualitative ionic-relaxation and alternate-placement benchmarks complete (Tables II and III; Table II 2/4 fixed-cell corners converged). Separate strain-only and doping-only workflows remain common, but total-energy effects under joint load are not necessarily additive. In quasi-hexagonal C₆₀ graphullerene we quantify the omitted cross term with the synergy order parameter $\mathcal{S}$ (meV per atom) [Eq. (1)]. On tetramers ($\sim 5\%$ nominal doping) and fifteen periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$, B/N/P @ $+3\%$, one substituent per C₆₀, ~ 1.67 at.%), tetramer α from the archived legacy PBE rigid grid (Table IV) is opposite for N (-303 meV/%) and B ($+57$ meV/%), whereas periodic PBE+D3 yields $|\mathcal{S}|$ up to 31.9 meV/atom at fixed coordinates for P at $n=1$ and $+3\%$ strain, with P showing the largest $|\mathcal{S}|$ despite the smallest tetramer $|\alpha|$. Sequential high-throughput screens can introduce systematic errors in the estimation of *relative* stability margins when local dopant distortion couples nonlinearly to external strain. Marcus reorganization energies and nearest-neighbor hopping parameters (Figs. 3 and 4) are reported for transport context but do not affect the non-additivity conclusions.

I. INTRODUCTION

Covalent molecular networks—from graphullerenes to covalent organic frameworks—are often tuned jointly by substitutional chemistry and mechanical load, yet many pipelines still estimate $E_{\text{tot}}(\epsilon, \delta)$ from *separate* doping-only and strain-only calculations [1–3]. Synthetic carbon allotropes with extended π conjugation underpin optoelectronic applications; charge transport in molecular networks is often polaronic and anisotropic [4–7], yet prior work emphasizes band edges, mobility, or hopping rather than a periodic total-energy cross term under joint biaxial (ϵ, δ) load (fifteen-point periodic audits here use PBE+D3; tetramer α in Table IV uses a separate legacy PBE archive). Quasi-hexagonal (qHP) C₆₀ graphullerene [4, 5] provides a covalently linked model network for such coupled tuning, with interfullerene linking that raises hopping couplings relative to vdW C₆₀ monolayers [4]; recent surveys of covalent fullerene networks emphasize synthesis, catalysis, and transport properties [8] but do not report a periodic total-energy cross term under joint biaxial (ϵ, δ) load. Ab initio models of pristine qHP networks report intrinsic intramolecular strain from the eight-bond hexagonal connectivity [9], so external biaxial load couples to an already distorted cage baseline. Beyond pristine monolayers, strain and substitution independently reshape band edges [2, 10, 11], while substitutional B/N engineering can yield spin-polarized half-semiconductor gaps [12], and related 2D fullerene networks show anisotropic mechanical and tribological response under load [13, 14] and endohedral qHP C₆₀ cage filling reshapes band gaps [15]; none of these reports a periodic total-energy cross term under joint biaxial (ϵ, δ) load, motivating explicit coupled grids rather than an additive superposition of composition and strain

effects. Non-additive coupling is documented in other two-dimensional networks [1, 16]; a quantitative *total-energy* synergy order parameter $\mathcal{S}$ [Eq. (1)] has not been reported for substitutional B/N/P qHP C₆₀ tetramers and $n = 1\text{--}8$ supercells; here, fifteen periodic (n, δ) points at $+3\%$ strain use PBE+D3, whereas tetramer α in Table IV comes from the archived legacy PBE rigid grid (no D3).

We address three questions on this model network: (1) how opposite linear strain coefficients α (meV per 1% biaxial strain of the *total* cell energy) emerge for donor vs. acceptor dopants; (2) how large the omitted synergy cross term $|\mathcal{S}|$ (meV/atom) is when strain and substitution are applied jointly; and (3) whether $|\mathcal{S}|$ tracks $|\alpha|$ or instead reflects local structural distortion. Using $\gtrsim 90$ converged CP2K tasks (Table I), we define $\mathcal{S}$ on periodic PBE+D3 grids (fifteen (n, δ) points at $+3\%$ strain) and report opposite donor/acceptor α on the archived tetramer strain grid (Table IV), demonstrating that the largest periodic $|\mathcal{S}|$ at fixed coordinates (up to 31.9 meV/atom for P at $n=1$) does not coincide with the smallest tetramer $|\alpha|$ for size-mismatched P. *Local* dopant distortion and *global* biaxial strain interact nonlinearly in total energy: $|\mathcal{S}|$ can exceed what linear α alone would suggest, so *relative* stability margins from $\epsilon = 0$ substitution energies or strain-only elastic increments need not survive a coupled load even if they do not reorder dopants with widely separated E_{sub} values. Reported $|\mathcal{S}|$ magnitudes serve as upper bounds—not predictions—at fixed coordinates until ionic-relaxation and alternate-placement benchmarks complete (Tables II and III; Table II 2/4 fixed-cell corners converged). Marcus λ and hopping observables (Figs. 3 and 4) characterize transport-relevant quantities but are not used to infer $\mathcal{S}$.

II. COMPUTATIONAL METHODS

A. Electronic-structure setup

All periodic audit and charged-state tasks use CP2K [17] with PBE [18] and Grimme DFT-D3 (Becke–Johnson damping) [19]; tetramer α and E_{sub} in Table IV come from the archived legacy PBE rigid grid (no D3). Table I summarizes basis sets, SCF convergence criteria, and the full task inventory. We do not employ the Koopmans-compliant rVV10 approach used in Ref. 4, and instead adopt a PBE+D3 protocol for periodic and charged-state tasks suitable for large-scale parameter scanning, at the cost of absolute accuracy in polaron binding energies.

B. Notation and conventions

Biaxial strain is denoted ϵ (percent); dopant species $\delta \in \{B, N, P\}$. The synergy order parameter is denoted by the calligraphic symbol $\mathcal{S}$ (never the Latin s), with units of meV per atom [Eq. (1)]. π -projected densities of states are written π -DOS (never bare “pi” in equations).

1. Rigid-strain protocol

Biaxial strain scales the in-plane lattice vectors of the *pristine* equilibrium cell at fixed fractional coordinates (no ionic relaxation under load). Reported α and $\mathcal{S}$ under this protocol are *upper bounds* at fixed coordinates, not equilibrium thermodynamic quantities. Table II benchmarks four fixed-cell geometry-optimization corners (pristine and P -substituted tetramer at $\epsilon=0$ and $+3\%$) to test whether $\mathcal{S}$ retains *sign* after ionic relaxation *within the same tetramer grid* (magnitude ratios are not quantitative until a matched-functional rigid reference exists); it does *not* validate periodic $n=4$ magnitudes, which differ in dopant concentration and boundary conditions. Main-text $|\mathcal{S}|$ values therefore serve as upper bounds—not predictions—of meV/atom-scale non-additivity until Tables II and III benchmarks complete (Sec. II E 3, Sec. II E 2).

2. Plane-wave cutoff (size scaling)

Archived CP2K inputs use 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (Table I); comparisons at fixed n always share the same cutoff. Cross- n trends therefore mix a modest plane-wave cutoff change at $n = 6$; we do *not* interpret the reported N $\mathcal{S}$ sign change between $n = 6$ and $n = 8$ as a physical size effect until the matched 400 Ry $n=6$ control in Table I completes.

C. Substitutional doping and grids

Tetramer models ($4 \times C_{60}$, 240 sites) substitute B, N, or P at nominal 5% (12–13 sites from archived coordinates); these dopant configurations are not thermodynamically optimized ground states. A second tetramer placement ensemble (18 single-point energy tasks; Table III) tests whether tetramer α and $\mathcal{S}$ are placement-sensitive at $\sim 5\%$ nominal doping. All periodic supercells propagate the same dopant site map (one substituent per C_{60}); quantitative $|\mathcal{S}|$ on the fifteen-point grid is therefore configuration-specific, whereas donor/acceptor sign trends for α on the reference tetramer placement are discussed as element-intrinsic qualitative expectations pending the alternate seed-137 map in Table III (Sec. II E 2). Periodic scaling cells place one substituent per C_{60} (n dopants among $60n$ lattice atoms, ~ 1.67 at.% for all n), contrasting with the tetramer grid ($\sim 5\%$ nominal). Counting a single dopant in the full supercell (e.g., $1/240=0.42$ at.% for the tetramer) is *not* the concentration used here; atomic percent is always dopants per lattice atom (n in $60n$). Tetramer α , gaps, and tetramer IPR/ J trends (Fig. 4) are therefore not directly interchangeable with periodic $\mathcal{S}(n)$ without a matched-concentration benchmark; all $\mathcal{S}$ claims use the periodic protocol. In total, $\gtrsim 90$ converged neutral tasks support the protocol-audited claims below (Table I); rigid-strain $|\mathcal{S}|$ magnitudes are upper bounds until Tables II and III complete. Charged-state geometry optimizations (UKS, multiplicity 2; SCF threshold $10^{-6} E_h$) supply adiabatic ionization potentials and electron affinities ($E_{\text{opt}}^{\pm}$). Eight vertical single-point energy evaluations compute $E_{\text{vert}}^{\pm}$ at the *neutral*-geometry coordinates with charged UKS settings matched to the relaxed state (SCF threshold $10^{-7} E_h$ by default; B $q = -1$: $10^{-6} E_h$), giving $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$ following Ref. 4. These Marcus diagnostics appear only in Fig. 3 and do not enter Eq. (1).

D. Energy observables

Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain, tetramer *total* energy) is obtained from the archived 24-point rigid tetramer grid (Table IV; legacy PBE without D3). Periodic $\mathcal{S}$ and size-scaling energies use PBE+D3 throughout (Table I). Positive α means tension *raises* total energy; negative α means tension *lowers* it under the rigid-strain protocol. Substitution energy $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is the total-energy difference per dopant site at fixed strain *without* elemental chemical-potential corrections; it is *not* a thermodynamic standard formation enthalpy ($\sim$ eV-scale tabulated values) and is used only for relative ranking within this protocol. In sequential high-throughput screens one often estimates $E_{\text{tot}}(\epsilon, \delta) \approx E_{\text{sub}}(\epsilon=0) + E_{\text{strain}}(\epsilon)$; the neglected term is precisely $\mathcal{S}(\epsilon, \delta)$ —the absolute error in that additive estimate. The order-of-magnitude ratio $|\mathcal{S}|/|E_{\text{sub}}|_{\text{per atom}}$ for screening workflows is discussed

in Sec. IV as a rough magnitude estimate only (not a core nonlinearity test).

1. Non-additive coupling metric

We quantify strain–doping cross terms with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times C_{60}$. The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} < 0$ indicates cooperative stabilization beyond the two-scan sum, whereas $\mathcal{S} > 0$ indicates cooperative destabilization.

When $\alpha_\delta = dE/d\epsilon$ denotes the *total*-cell linear strain coefficient (meV per 1% strain) at fixed n , a first-order estimate of the per-atom cross term is $\mathcal{S}(\epsilon, \delta) \approx (\alpha_\delta - \alpha_0)\epsilon/N_{\text{atoms}}$ at small ϵ under the rigid-strain protocol; tetramer and periodic n use different N_{atoms} and dopant concentrations, so this link is not used quantitatively in the periodic audit (Discussion, argument scope).

Periodic-cell $\mathcal{S}$ values are extracted from converged CP2K total energies (Table I); the fifteen-point +3% strain audit accompanies the open data release. We report $\mathcal{S}$ for total energy only; an analogous gap cross term $\mathcal{S}_{\text{gap}}$ is not computed here—main-text panel (a) shows that gap slopes need not track $|\mathcal{S}|$.

2. Localization and coupling (Fig. 4)

Following Capobianco *et al.* [4], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling J from tetramer dimer fragments. IPR is $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$ on a Mulliken-projected orbital grid; verified values for the four corners in Table I accompany the open data release. J is estimated from the ground–first-excited splitting of the anionic dimer at fixed interfullerene geometry (two-state model) and plotted in Fig. 4. These observables use the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant $J \simeq 75\text{--}81$ meV reported for pristine qHP networks [4], but they provide an internally consistent trend under coupled (ϵ, δ) .

3. Polaron binding

Marcus reorganization energies follow the vertical–adiabatic decomposition [4]: $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$, with separate internal ($\lambda_{\text{loc}}^\pm$) and environmental ($\lambda_{\text{env}}^\pm$) contributions when geometry subsets are relaxed. Charged-state

relaxations and vertical single-points are documented in Fig. 3. Charged geometry optimizations supply adiabatic IP/EA; vertical single-points at the neutral geometry yield Marcus $\lambda^\pm$ (Fig. 3).

E. Detailed calculation protocols

1. DFT calculator settings

Periodic total-energy and electronic-structure tasks use CP2K [17] with PBE [18] and Grimme DFT-D3 (Becke–Johnson damping) [19]. Tetramer α and E_{sub} in Table IV come from the archived rigid tetramer grid (legacy PBE without D3). Unless stated otherwise, biaxial strain scales the in-plane lattice vectors of the pristine equilibrium cell at fixed fractional coordinates (rigid-strain protocol). The plane-wave cutoff is 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (Table I); the atom-centered basis (DZVP-MOLOPT-SR-GTH) is unchanged across n . Tetramer models ($4 \times C_{60}$, 240 atoms) use B/N/P substitution at nominal 5% (12–13 dopant sites from archived coordinates). Periodic supercells place one substituent per C_{60} unit ($n = 1\text{--}8$), giving n dopants among $60n$ lattice atoms (~ 1.67 at.% for all n).

2. Alternate-placement benchmark (Table III)

The *reference* placement in Table III is the archived rigid tetramer map underlying Table IV (legacy PBE without D3). The *alternate* placement repeats the full B/N/P strain grid ($\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$; 18 rigid single-point tasks) with an independently rerolled substitution map at the same nominal 5% doping level (seed 137). Reference-placement $\mathcal{S}_{+3\%}^{\text{tet}}$ for B/N/P is tabulated in Table III and Sec. III C; periodic $\mathcal{S}(n)$ is not inferred from either tetramer map. Alternate tasks use PBE+D3 rigid single-point total-energy evaluations at 400 Ry with an SCF convergence threshold of $10^{-6} E_h$; they are *not* on the same exchange-correlation treatment as the reference α column, so only within-placement trends are compared once the grid converges.

3. Ionic-relaxation benchmark (Table II)

The four tetramer corners in Table II are pristine and P -substituted cells at $\epsilon=0$ and +3% biaxial strain (the Eq. (1) grid for P). They use fixed-cell geometry optimization with BFGS (maximum 300 ionic steps) at prescribed in-plane cell vectors; only fractional coordinates relax while lattice vectors remain fixed. BFGS stops at 300 steps or when the maximum and RMS force components fall below 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr, or when the maximum and RMS atomic displacements fall below 3.0×10^{-3} and 1.5×10^{-3} bohr, respectively. These tasks use the same PBE+D3 settings as periodic audits:

400 Ry plane-wave cutoff, DZVP-MOLOPT-SR-GTH basis, and an SCF convergence threshold of $10^{-6} E_h$ with OT-DIIS and up to 20 outer SCF cycles. Rigid reference energies in Table II remain the archived legacy-PBE (no D3) single-point grid; relaxed and rigid corners therefore differ in exchange-correlation treatment, and only sign retention of $\mathcal{S}$ is interpreted.

4. Marcus vertical step and post-processing

Charged-state geometry optimizations supply adiabatic ionization potentials and electron affinities ($E_{\text{opt}}^{\pm}$). Eight vertical single-point energy evaluations compute $E_{\text{vert}}^{\pm}$ at the *neutral*-geometry coordinates with the charged UKS settings of the relaxed state, following Ref. 4; $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$ (Fig. 3). Figure 3 schematically separates adiabatic IP (slanted arrow between neutral and charged minima) from the neutral-geometry vertical single-points that define $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$; adiabatic EA is quoted as text on the B and P panels only. π -projected densities of states use a 0.05 eV Gaussian broadening aligned to the Fermi level (Fig. 2). Inverse participation ratios and hopping integrals J are defined in Sec. II; verified IPR values accompany the open data release, whereas Fig. 4 reports J only on the tetramer dimer grid.

F. Validation benchmarks

The full calculation map (converged tasks, main-text usage, and pending controls) is in Table I; ionic-relaxation energies (Table II) and the alternate-placement grid (Table III) appear there when complete. Three validation benchmarks—rigid vs. relaxed tetramer $\mathcal{S}_P$ (Table II), alternate dopant placement (Table III; reference-placement $\mathcal{S}_{+3\%}^{\text{tet}}$ tabulated in Sec. III C; alternate grid PBE+D3 on seed 137, Sec. II E 2), and a matched 400 Ry $n=6$ N control (Table I)—govern interpretation of the N $\mathcal{S}(n)$ sign change cited in Secs. III C and IV. Table II has completed the pristine fixed-cell geometry-optimization corners at $\epsilon=0$ and $+3\%$ ($E = -1363.9551$ and $-1363.9550 E_h$); $\mathcal{S}_{\text{relaxed}}$ and sign retention await the two P-doped fixed-cell corners (Sec. II E 3).

Table IV and Sec. III B report E_{sub} using the substitution-energy definition above (no chemical-potential reference; not a formation enthalpy).

III. RESULTS

We organize the results around total-energy non-additivity under the rigid-strain upper-bound protocol (Sec. II B); Tables II and III gate relaxation- and placement-sensitive interpretations. Sec. III A summarizes band-edge context; Sec. III B reports linear strain

coefficients, E_{sub} , and geometric strain response (Table V); Sec. III C gives $\mathcal{S}(n)$ at $+3\%$ strain across dopants and supercell sizes (Table I; fifteen-point periodic audit); Sec. III D reports Marcus λ and IPR/J after the $\mathcal{S}$ audit.

A. Electronic structure under strain and doping

Figure 1(a,b) summarizes tetramer HOMO–LUMO gaps (archived tetramer electronic-structure set) and representative π -DOS (B @ $\epsilon=0$; color scheme follows Ref. 4). At $\epsilon=0$, B is near a closed π gap ($E_g \approx 0.03$ eV), N doping pins the Fermi level inside the conduction band ($E_g \approx -0.14$ eV), and P remains gapped ($E_g \approx 0.05$ eV); tensile strain narrows the pristine gap (~ 23 meV/% on the three available strain points in the dataset). Donor vs. acceptor ordering of the $\epsilon=0$ gaps tracks the opposite α signs in Sec. III B, but band-edge slopes are not used to infer $\mathcal{S}$. The four-panel π -DOS stack (Fig. 2) shows N-induced states crossing the Fermi level at $\epsilon=0$ (metallic pinning), a narrow π gap for B ($E_g \approx 0.03$ eV), and a wider gap for P—consistent with excess vs. deficient π electron count at the dopant cage but not yet decomposed into site-resolved Mulliken or Bader partitions (Limitations).

B. Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/%). On the tetramer grid (Table IV), substitution opens an order-of-magnitude spread in α : N (-303.2 meV/%, tension stabilizes the doped cell), B ($+56.9$ meV/%, tension destabilizes), and P (-19.1 meV/%, weakly stabilizing). The opposite sign for N vs. B persists across the $\epsilon \in \{-5, \dots, +5\}\%$ grid at this fixed placement and is not an artifact of a single strain point; Table III will test whether the slopes are placement-sensitive. At $\epsilon = +5\%$, N remains the most tension-stabilized dopant (α most negative), whereas B shows the strongest destabilization under tension; P sits between these extremes with a modest negative α , foreshadowing why its linear coefficient poorly predicts $|\mathcal{S}| = 23.7$ meV/atom for P at $n=4$ under $+3\%$ strain (Sec. III C). Substitution energies at $\epsilon = 0$ span $E_{\text{sub}} = +73.4$ eV/dopant (B), -115.3 eV/dopant (N), and -15.8 eV/dopant (P)—raw total-energy differences without chemical-potential reference, not standard formation enthalpies—with $n_{\text{dop}} = 12\text{--}13$ counted from input coordinates (Table IV). Figure 1(c) highlights the α split; ranking dopants solely by $|E_{\text{sub}}|$ at $\epsilon=0$ is misleading once $+3\%$ strain is applied. Table V links this mechanical spread to local geometry: P shows negligible strain-induced change in mean nearest C distance ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å vs. $\sim 8.5 \times 10^{-3}$ Å for B at $+3\%$ load), foreshadowing the α – $\mathcal{S}$ decoupling quantified in Sec. III C (Discussion mechanistic synthesis).

TABLE I. Summary of converged DFT calculations and correspondence to main-text results. The ~ 90 tasks cited in the main text are the converged calculations that directly support quantitative claims; the full archive includes additional preliminary grids (rows marked “not used”).

Calculation set	Model	Converged tasks	Main-text usage	Notes
Early strain probe	$2 \times C_{60}$	5	not used	superseded by tetramer grids
Single-molecule doping	C_{60}	11	not used	superseded
Early electronics	$2 \times C_{60}$	24	not used	superseded by tetramer π -DOS
Polaron IPR (tetramer dimer)	tetramer	4	open data release	pristine/B $\times \epsilon=0, +3\%$
Nearest-neighbor coupling J	tetramer dimer	4	Fig. 4	same four corners (PBE+D3)
Tetramer strain energies	tetramer	24	Table IV, α	legacy PBE rigid; no D3
Optimal-design grid	tetramer	20	not used	archive only
Tetramer π -DOS/gaps	tetramer	12	Fig. 2; main panels	
Geometry references	tetramer relax	6	charged-state inputs	reference geometries
Charged polarons	tetramer $q = \pm 1$	12 geometry opt.	Fig. 3	Marcus $\lambda^\pm$ and adiabatic IP/EA
Size scaling	$n \times C_{60}$	40 (+1 pending)	main $\mathcal{S}(n)$	400 Ry ($n \leq 4$), 350 Ry ($n \geq 6$); cutoff400 control
$n=6$ cutoff control	$6 \times C_{60}$ N @+3%	pending	N sign benchmark	400 Ry matched control (pending)
Ionic relaxation (tetramer)	tetramer	2/4 geometry opt.	Table II	fixed-cell PBE+D3 geometry optimization (BFGS)

TABLE II. Ionic-relaxation benchmark for the tetramer P -substitution synergy grid (four fixed-cell geometry-optimization corners: pristine and P at $\epsilon=0$ and $+3\%$, same biaxial cell). This benchmark is *sign-qualitative only*: the rigid reference uses legacy PBE without D3, whereas relaxed corners use PBE+D3, so $|S_{\text{relaxed}}|/|S_{\text{rigid}}|$ is *not* a quantitative retention ratio until a matched-functional rigid grid is recomputed. We test whether $\mathcal{S}$ retains *sign within the tetramer grid only*; it does *not* calibrate periodic main-text $\mathcal{S}(n)$, which uses different dopant concentration and boundary conditions. Rigid reference energies: archived fixed-coordinate tetramer single-point grid (legacy PBE, no D3). Relaxed energies: PBE+D3 fixed-cell geometry optimization (400 Ry) from archived inputs in the open repository. $\mathcal{S}$ follows Eq. (1) with $N=240$ atoms.

Quantity	Rigid (legacy tetramer)	Relaxed (geometry opt.)
$E(\text{pristine}, \epsilon=0) (E_h)$	-1363.5512	-1363.9551
$E(\text{pristine}, \epsilon=3\%) (E_h)$	-1363.5522	-1363.9550
$E(P, \epsilon=0) (E_h)$	-1370.4965	pending
$E(P, \epsilon=3\%) (E_h)$	-1370.4890	pending
$\mathcal{S}$ (meV/atom)	+0.96	pending
Sign of $\mathcal{S}$ vs. rigid	pending (2/4 geometry opt.; P -doped fixed-cell corners in progress)	

C. Non-additive strain–doping coupling

Figure 1(d) tracks protocol $\mathcal{S}(n)$ (rigid-strain upper bounds) with one dopant per C_{60} at $+3\%$ biaxial strain. At $n = 4$ under the fixed periodic dopant map (reference tetramer placement in Table III; alternate seed-137 grid pending, Sec. II E 2), $\mathcal{S} = -6.2$ meV/atom (B), $+3.0$ meV/atom (N), and -23.7 meV/atom (P); P shows the largest magnitude despite the smallest $|\alpha|$ on the tetramer grid (Table IV). Across fifteen periodic points ($n \in \{1, 2, 4, 6, 8\}$, B/N/P), $|\mathcal{S}|$ reaches 31.9 meV/atom for P at $n = 1$ and remains $\mathcal{O}(10)$ meV/atom at larger n for B and P, whereas N is smaller in magnitude at $n = 4$ but comparable at $n = 1$. For N, $\mathcal{S}$ changes sign between $n = 6$ ($+2.4$ meV/atom) and $n = 8$ (-2.3 meV/atom) at the production 350 Ry cutoff; this reversal is *not* used in the main conclusions until the matched 400 Ry $n=6$ control in Table I and Tables II and III complete (Sec. II F). Core size-scaling claims focus on $n \leq 4$ at matched 400 Ry cutoffs; B and P show monotonic $|\mathcal{S}|$ decrease from $n = 1$ toward $n = 4$, whereas $n \geq 6$ trends are reported for completeness only and mix the 350 Ry production cutoff. We report provisional $\mathcal{S}(n) \approx \mathcal{S}_\infty + A/n$

fits in the open data release but refrain from drawing conclusions about the asymptotic $\mathcal{S}_\infty$ value until cutoffs are matched across n . All periodic $\mathcal{S}$ values quoted in this subsection use the fixed dopant-site map and are tabulated in Table I (fifteen $n \times$ dopant points at $+3\%$ strain). On the reference tetramer placement, legacy PBE rigid $\mathcal{S}_{+3\%}^{\text{tet}}$ from Eq. (1) is $+3.8$, $+5.7$, and $+1.0$ meV/atom for B, N, and P (Table III); the pending alternate-placement grid uses PBE+D3 rigid single-points on an independent site map (Sec. II E 2) and is not used to infer periodic $\mathcal{S}(n)$. Negative $\mathcal{S}$ (B, P at $n = 4$) means coupled tension *stabilizes* the doped cell relative to the additive estimate; positive $\mathcal{S}$ (N at $n = 4$) means sequential screens *overestimate* stability under load.

1. Ionic-relaxation checkpoint (partial)

Table II benchmarks four fixed-cell geometry-optimization corners on the tetramer P -substitution grid (pristine and P at $\epsilon=0$ and $+3\%$); pristine corners at both strains have converged (2/4), and the two P -doped corners remain in progress. For the pristine tetramer, re-

TABLE III. Alternate dopant-placement sensitivity (*tetramer only*, rigid strain). The *reference* placement defines main-text tetramer α (Table IV) and the periodic supercell site map (one substituent per C_{60}); its α and $\mathcal{S}_{+3\%}^{\text{tet}}$ use the archived legacy-PBE grid (no D3). The *alternate* placement repeats the full B/N/P strain grid ($\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$; 18 rigid single-points) on an independent seed-137 site map with PBE+D3 (Sec. II E 2); pending rows test placement sensitivity at $\sim 5\%$ nominal doping without mixing periodic boundary conditions. $\mathcal{S}_{+3\%}^{\text{tet}}$ follows Eq. (1) on the archived rigid tetramer four-corner grid ($N=240$). Periodic $\mathcal{S}(n)$ is *not* tabulated here because concentration and boundary conditions differ; main-text periodic magnitudes are configuration-specific until alternate periodic placements are computed.

Dopant	Placement	α (meV/%)	$\mathcal{S}_{+3\%}^{\text{tet}}$ (meV/atom)	Status
B	reference	+56.9	+3.8	verified (Table IV)
N	reference	-303.2	+5.7	verified
P	reference	-19.1	+1.0	verified
B	alternate	pending	pending	pending
N	alternate	pending	pending	pending
P	alternate	pending	pending	pending

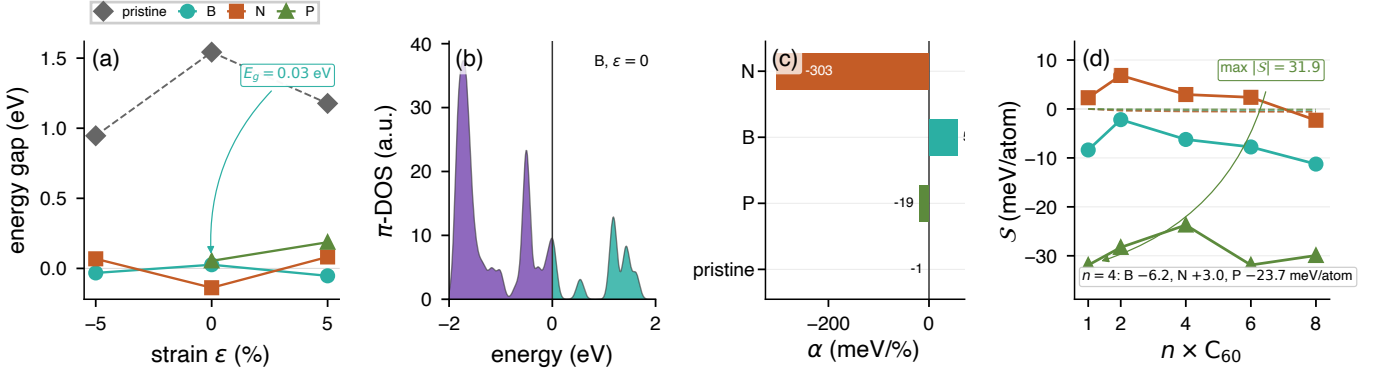


FIG. 1. Coupled (ϵ, δ) response in doped qHP C_{60} (PBE+D3). (a) HOMO-LUMO gap vs. strain (E_g callout: B @ $\epsilon=0$). (b) π -DOS for B @ $\epsilon=0$ (color scheme follows Ref. 4). (c) Mechanical α (Table IV; legacy PBE rigid grid). (d) $\mathcal{S}(n)$ at +3% (peak $|\mathcal{S}|$ at P, $n=1$; inset lists $n=4$ values for B/N/P); Marcus λ in Fig. 3 and hopping J in Fig. 4 (IPR tabulated in the open data release).

laxed total energies at $\epsilon=0$ and +3% are -1363.9551 and $-1363.9550 E_h$ ($\Delta E \approx 1.2 \times 10^{-4} E_h$), roughly an order of magnitude smaller than the archived rigid legacy-PBE nominal strain increment ($\sim 10^{-3} E_h$). Because rigid references use legacy PBE without D3 whereas relaxed corners use PBE+D3 (Sec. II E 3), only *sign* retention of $\mathcal{S}$ is interpreted once all four corners complete. $\mathcal{S}_{\text{relaxed}}$ and sign comparison to the rigid +0.96 meV/atom entry await the two P-doped fixed-cell corners; Sec. IV links this interim pristine strain-energy shrinkage to the upper-bound reading of load-dependent margins.

D. Polaron transport diagnostics

Figure 3 reports Marcus reorganization energies from twelve charged-state geometry optimizations and eight vertical single-points at the neutral geometry (Sec. II E 4). Columns compare the *same* Marcus workflow across dopants: the slanted arrow is adiabatic IP, teal text gives adiabatic EA on B/P, and boxed $\lambda^\pm$ values come from vertical minus adiabatic energies at the neutral geometry.

On the tetramer dimer grid, B substitution at $\epsilon=0$ delocalizes the charged-state orbital (IPR $75.0 \rightarrow 24.6$) while raising J from 26.6 to 46.4 meV; coupled B @ +3% gives $J = 37.2$ meV and IPR 23.1. The criterion $J > \lambda/2$ for polaron hopping dominance [4] is *not* met at PBE+D3 ($J < \lambda/2$), so a polaron transition is not confirmed on this grid.

IV. DISCUSSION

A. Literature context

Relative to strain-only, doping-only, mobility, or mechanical studies in qHP and related fullerene networks [1, 2, 4, 7–15, 20, 21], the distinction here is a periodic *total-energy* audit of joint biaxial (ϵ, δ) load [Eq. (1)] on a rigid-strain upper-bound protocol (periodic PBE+D3; tetramer α from legacy PBE, Table IV). Strain-engineering roadmaps for generic two-dimensional materials [3] likewise foreground band-edge and mobility levers under separate mechanical or chemical protocols rather than a matched four-corner total-

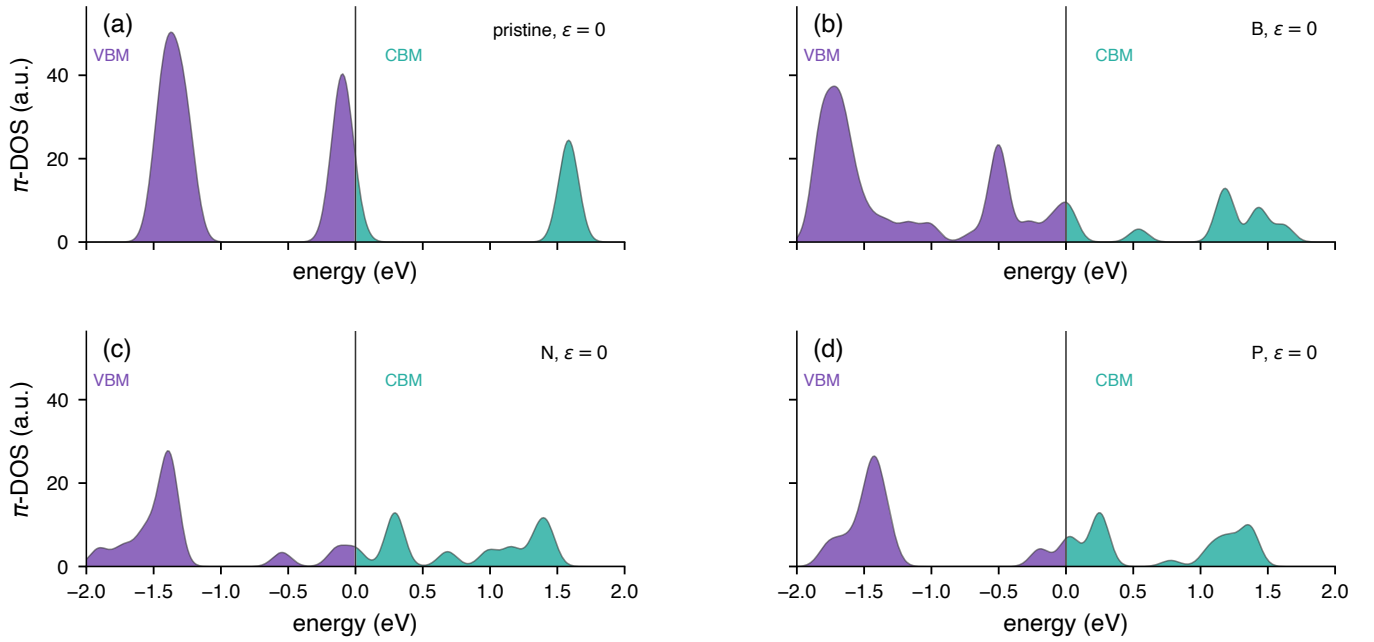


FIG. 2. π -projected density of states (π -DOS) for pristine and B/N/P-doped tetramers at $\epsilon=0$ (PBE+D3). Panels (a)–(d) share a common vertical scale; energy is relative to the Fermi level at 0 eV (Capobianco color scheme [4]).

TABLE IV. Linear strain sensitivities and substitution energies for tetramer models (archived rigid-strain grid, legacy PBE without D3; not mixed with periodic PBE+D3 $\mathcal{S}$ in Table I). $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at $\epsilon=0$ with no chemical-potential reference (not a standard formation enthalpy). n_{dopants} is counted from archived input coordinates (nominal 5% target; B: 13, N: 12, P: 12).

System	n_{dop}	$E(\epsilon=0) (E_h)$	$E(\epsilon=+5\%) (E_h)$	α (meV/%)	E_{sub} (eV/dopant)
Pristine	—	−1363.5512	−1363.5509	−1.0	—
B-doped	13	−1328.4900	−1328.4090	+56.9	+73.4
N-doped	12	−1414.3812	−1414.4053	−303.2	−115.3
P-doped	12	−1370.4965	−1370.4888	−19.1	−15.8

energy grid on covalent fullerene networks. Graphendofullerene derivatives report Curie temperatures tunable through *either* biaxial strain or electrostatic doping applied in isolation [11], reinforcing that joint (ϵ, δ) total-energy corners are not implicit in sequential protocol designs. In other 2D hosts, substitutional defects can raise or lower formation energies under biaxial strain depending on dopant–host size mismatch [21]; spin-resolved impurity engineering on qHP graphullerene instead targets half-semiconductor gaps [12]; isomer-engineered C_{70} network designs similarly tune band edges through topology choice rather than joint substitutional–mechanical load [22]; we instead quantify the omitted additive cross term $\mathcal{S}$ in total energy. Quasi-harmonic thermal-expansion studies of qHP monolayers report anisotropic positive and negative lattice response tied to interfullerene bond topology [10, 20], whereas our audit holds fractional coordinates fixed at zero-temperature total energy (upper bound; partial ionic-relaxation and alternate-placement checkpoints in Secs. IIIC and IIF and Tables II and III). Strain-engineering studies on

smaller fullerene graphullerene instead emphasize magnetic or half-metallic transitions [23], which is orthogonal to the neutral total-energy audit pursued here. Band-edge, spin-resolved, and hopping metrics (Fig. 4) contextualize mechanism but are not used to infer $\mathcal{S}$.

B. Mechanistic synthesis

a. Opposite α for N vs. B. N introduces excess π electrons that strengthen bonding under tension (negative α), whereas B creates electron deficiency and antibonding π^* character that rises with lattice expansion (positive α). Quantitatively, the $\epsilon=0$ gap ordering ($E_g^N < E_g^B < E_g^P$ within PBE; Sec. III A) and the dopant-resolved π -DOS stack in Figs. 1(b) and 2 (N states at E_F ; B near-gap; P gapped) align with donor vs. acceptor π counting, while total-energy α spans ~ 360 meV/% between N and B on the tetramer grid (Table IV). The opposite N vs. B sign is reported for this fixed placement across $\epsilon \in \{-5, \dots, +5\}\%$ (Table IV); Table III

TABLE V. Local dopant–cage structure from archived tetramer XYZ (mean nearest C distance $\bar{d}$ in Å; standard deviation $\sigma(\bar{d})$ across dopant sites; covalent-radius excess vs. C in pm). Strain-induced change $\Delta\bar{d} = \bar{d}(\epsilon = +3\%) - \bar{d}(\epsilon=0)$ summarizes how tensile load couples to pre-existing distortion (Discussion mechanistic synthesis (ii),(vi)). Values are computed from archived tetramer XYZ coordinates in the open data release; Mayer bond order and Bader charges are not included.

Dopant	$\bar{d}(\epsilon=0)$ (Å)	$\bar{d}(\epsilon = +3\%)$ (Å)	$\Delta\bar{d}$ (Å)	$\sigma(\bar{d})$ @ +3% (Å)	Δr_{cov} (pm)
B	1.384	1.393	+0.0085	0.019	+7
N	1.393	1.402	+0.0087	0.024	−6
P	1.384	1.385	+0.00003	0.0004	+30

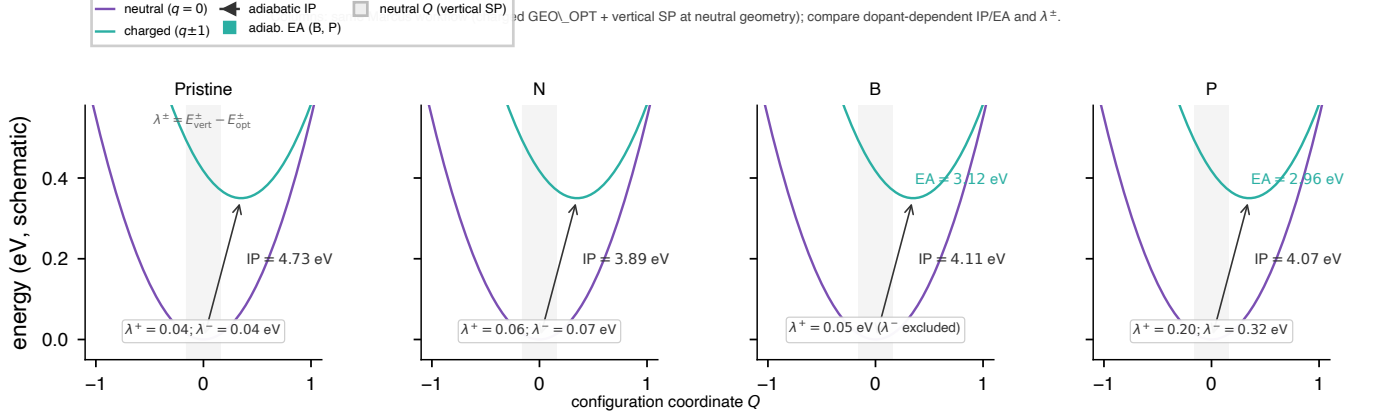


FIG. 3. Marcus reorganization energies across dopants (12/12 charged-state geometry optimizations; eight vertical single-points at the neutral geometry). Each column applies the same protocol; the slanted arrow marks adiabatic IP ($q=+1$ GEO.OPT), teal labels give adiabatic EA where computed (B, P), and the shaded $Q \approx 0$ band marks the neutral geometry used for vertical single-points ($\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$). Pristine $\lambda^\pm = 0.041/0.042$ eV; N $\lambda^\pm = 0.058/0.065$ eV; B $\lambda^+ = 0.051$ eV; P $\lambda^+ = 0.200/\lambda^- = 0.315$ eV. The electron-channel λ^- for B is unphysical under the fixed neutral-geometry vertical protocol and is excluded (Limitations).

will repeat the full tetramer strain grid on an alternate seed-137 site map (PBE+D3 rigid; Sec. II E 2) to test slope sensitivity at $\sim 5\%$ nominal doping without mixing periodic boundary conditions. Mayer bond order and Bader charge partitioning along the strain path are not yet computed (Limitations); periodic $\mathcal{S}(n)$ magnitudes are likewise not averaged over alternate supercell placements until such audits exist.

b. Large $|\mathcal{S}|$ for P despite small $|\alpha|$. P carries the largest covalent-radius excess (+30 pm vs. C; Table V) and pre-existing cage distortion at $\epsilon=0$, yet shows the smallest strain-induced change in mean nearest C distance ($\Delta\bar{d} \approx 3 \times 10^{-5}$ Å vs. $\sim 8.5 \times 10^{-3}$ Å for B/N) and the smallest site-to-site spread at +3% ($\sigma(\bar{d}) \approx 4 \times 10^{-4}$ Å vs. $\sim 2 \times 10^{-2}$ Å for B/N; Table V). Tensile load therefore couples to *pre-existing* cage distortion from size mismatch rather than to a uniform elastic response, yielding a cross term invisible in separate scans but dominant when both strain and substitution are applied simultaneously—consistent with P’s intermediate E_{sub} yet the largest $|\mathcal{S}|$ at $n=4$ (23.7 meV/atom; Table I) despite weak tetramer $|\alpha|$. Sec. III B presents this Table V geometric decoupling before the $n=4$ $\mathcal{S}$ ranking in Sec. III C: tetramer $|\alpha|$ ranks P as weakly strain-sensitive, whereas the coupled periodic audit exposes the

largest cross term.

c. N $\mathcal{S}$ sign change with n . With one substituent per C_{60} , electrostatic fields dilute at large n . The tetramer linear coefficient α and the periodic cross term $\mathcal{S}$ probe different energy partitions: α is the slope of *total-cell* energy under uniform lattice expansion at $\sim 5\%$ multi-site doping, whereas $\mathcal{S}$ is the omitted four-corner cross term per atom at ~ 1.67 at.% under periodic boundary conditions. They therefore need not share sign even for the same element—a negative α (tension lowers total energy in a strain-only scan) can coexist with positive $\mathcal{S}$ (the coupled corner is *less* stable than an additive model predicts), as for N at $n=4$. Any N sign flip between $n=6$ and $n=8$ is additionally confounded with the 400→350 Ry cutoff step and remains uninterpreted (Sec. II F). Together, these three classes suggest a taxonomy for covalent molecular networks: *electronic* donor/acceptor control of α , *structural* local-distortion coupling to external strain (P), and *finite-size* electrostatic dilution (N at large n).

d. Geometric vs. electronic nonlinearity. For P, Table V shows that rigid tensile load barely moves the mean dopant–C distance ($\Delta\bar{d} \approx 3 \times 10^{-5}$ Å) yet $\mathcal{S}$ remains the largest at $n=4$ (23.7 meV/atom), whereas B/N exhibit $\sim 10^2 \times$ larger $\Delta\bar{d}$ at +3% but smaller $|\mathcal{S}|$ (6.2 and

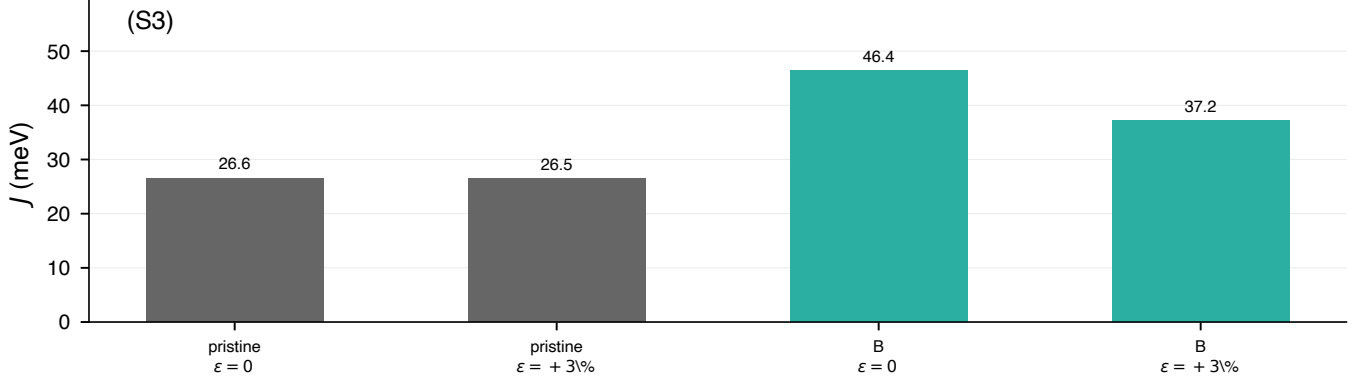


FIG. 4. Nearest-neighbor electronic coupling J on the tetramer dimer grid (pristine and B-doped cells at $\epsilon=0$ and $+3\%$; PBE+D3). Companion inverse-participation-ratio values for the same four corners are tabulated in the open data release (Methods).

3.0 meV/atom). At $n=4$ the coupled correction for P exceeds the rigid per-atom strain increment from tetramer α by two orders of magnitude ($|\mathcal{S}| \gg \alpha \times 3\%/N_{\text{atoms}}$; Methods energy observables), even though $|\alpha|$ ranks P as the weakest strain responder on the tetramer grid.

e. Argument scope (concentration and boundary conditions). We do *not* validate $\mathcal{S}$ by extrapolating tetramer α to periodic $n=4$: the $\sim 5\%$ tetramer vs. $\sim 1.67\%$ periodic concentrations and tetramer vs. periodic boundary conditions are unmatched, so a first-order $(\alpha_\delta - \alpha_0)\epsilon/N_{\text{atoms}}$ estimate is excluded from the argument. Non-additivity is assessed only through Eq. (1) on matched four-corner grids at common (ϵ, δ) ; P in particular requires explicit coupled evaluation rather than sequential α screening.

f. Decomposition outlook. The Table V decoupling indicates that the cross term is not reducible to uniform bond stretching alone; electronic rehybridization and pre-stored cage strain likely contribute, but their relative weights are not partitioned here without Mayer/Bader analysis along the strain path (Limitations).

g. Significance versus DFT uncertainty. $\mathcal{S}$ is a difference of total energies computed with the *same* functional, basis, and cell for all four corners of Eq. (1), so systematic errors largely cancel in sign and relative magnitude. As a *rough magnitude estimate only* (not a core result), at $n=4$ and $+3\%$ strain P has $|\mathcal{S}| = 23.7$ meV/atom while the tetramer P per-atom substitution scale is $|E_{\text{sub}}|n_{\text{dop}}/n_{\text{atoms}} \approx 0.79$ eV/atom ($|E_{\text{sub}}|=15.8$ eV/dopant, $n_{\text{dop}}=12$ on the 240-atom tetramer, Table IV)—i.e. $|\mathcal{S}|/|E_{\text{sub}}|_{\text{per atom}} \sim 3\%$ —but tetramer ($\sim 5\%$) and periodic $n=4$ ($\sim 1.67\%$) concentrations differ and Table IV is not on the same functional as periodic $\mathcal{S}$. Large absolute uncertainties in PBE E_{sub} therefore do not invalidate the conclusion that sequential scans omit a meV/atom-scale cross term.

C. Design implications

At $n=4$ and $+3\%$ strain the coupled correction is $|\mathcal{S}| = 23.7$ meV/atom for P vs. 6.2 meV/atom for B—sequential *additive* estimates differ by approximately a factor of four in the load-dependent *margin* even though widely separated E_{sub} ranks at $\epsilon=0$ (e.g., B vs. P; Table IV) are unchanged. Dopants with larger covalent-radius excess (P: $+30$ pm vs. B: $+7$ pm; Table V) show the largest protocol $|\mathcal{S}|$ under fixed coordinates; *size-mismatched* substituents may warrant coupled (ϵ, δ) evaluation rather than one-parameter pipelines once Tables II and III benchmarks complete. Interim pristine corners on the tetramer P -substitution four-corner grid (Table II; pristine $\epsilon=0$ and $+3\%$ converged, P -doped corners in progress; Sec. IIIC) shrink the nominal tensile strain increment by roughly an order of magnitude relative to the rigid legacy reference, supporting the upper-bound reading of protocol $|\mathcal{S}|$ on load-dependent margins before the P -doped corners complete. The fifteen-point periodic grid tabulates protocol $|\mathcal{S}|$ at each (n, δ) as an audit of additive omission under the rigid protocol (Table I), not as quantitative design targets. At $+3\%$ strain, P carries the largest additive-approximation error in protocol $\mathcal{S}$ ($|\mathcal{S}| = 23.7\text{--}31.9$ meV/atom from $n=4$ to $n=1$ under fixed coordinates); qualitative co-design should treat P under tension as a *coupled* stability problem pending Tables II and III, not $|E_{\text{sub}}|$ at $\epsilon = 0$ alone. Where $\mathcal{S} < 0$ (B and P at $n=4$), coupled tension can stabilize doped cells beyond additive predictions; where $\mathcal{S} > 0$ (N at $n=4$), sequential screens overestimate stability under load. For chemistries with comparable $|E_{\text{sub}}|$ or in multi-dopant competition, omitting $\mathcal{S}$ can shift margins by several percent of the per-atom substitution scale—much larger than the rigid strain increment from α —even when the global dopant rank at $\epsilon=0$ is unchanged. Gap closing [Fig. 1(a)] does not track $|\mathcal{S}|$; we do not compute $\mathcal{S}_{\text{gap}}$, so photonic or transport targets cannot be inferred from total-energy $\mathcal{S}$ alone. For B at $n = 4$, both negative

$\mathcal{S}$ and positive α imply that a strain-only elastic estimate would predict tension *raises* energy while the coupled scan *lowers* it relative to the additive model—a direct sign conflict that sequential pipelines cannot resolve without evaluating $\mathcal{S}$. Physically, B’s positive α reflects acceptor-induced π^* antibonding that rises with lattice expansion, whereas negative $\mathcal{S}$ means the *joint* (ϵ, δ) corner is more stable than $E(\epsilon, 0) + E(0, \delta) - E(0, 0)$ would predict: local cage relaxation pathways and dopant–network hybridization redistribute strain energy beyond the linear slope. For N at $n=4$, negative tetramer α and positive $\mathcal{S}$ present a second sign conflict: tension stabilizes the doped tetramer in a strain-only scan, yet the periodic coupled audit reports a positive cross term that sequential additive models fail to capture at ~ 1.67 at.% doping because the four-corner $\mathcal{S}$ measures cooperative *destabilization* relative to separate scans, not the slope of a single-branch strain curve. N $\mathcal{S}$ vs. n is reported for completeness but excluded from design rules until Tables I–III controls converge.

D. Validation protocol (Tables I–III)

Three validation benchmarks characterize protocol limits without altering the periodic magnitudes quoted above (Sec. III C reports the interim Table II pristine checkpoint).

a. Ionic-relaxation benchmark (Table II). Fixed-cell geometry optimization on the four Eq. (1) corners for the tetramer P -substitution grid (pristine and P at $\epsilon=0$ and $+3\%$); because rigid references use legacy PBE without D3 whereas relaxed corners use PBE+D3, the acceptance criterion is *sign retention* of $\mathcal{S}$ after ionic relaxation within that tetramer grid—not a quantitative $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}|$ ratio until a matched-functional rigid grid is recomputed. Interim pristine corners (Sec. III C) already show $\Delta E \approx 1.2 \times 10^{-4} E_h$ after ionic relaxation versus $\sim 10^{-3} E_h$ on the rigid legacy reference, so rigid-strain $|\mathcal{S}|$ should be read as an upper bound on the strain increment until the P-doped corners complete; a matched-functional rigid grid would be needed before interpreting any magnitude ratio. It does *not* rescale periodic $\mathcal{S}(n)$ because concentration and boundary conditions differ.

b. Alternate-placement benchmark (Table III). The reference map supplies archived legacy-PBE α and $\mathcal{S}_{+3\%}^{\text{tet}}$ (Table IV); the pending alternate map repeats the strain grid with PBE+D3 rigid single-points on an independent seed-137 site map (Sec. II E 2), so only within-placement trends are compared once converged. Reference-placement $\mathcal{S}_{+3\%}^{\text{tet}}$ is already tabulated for B, N, and P ($+3.8$, $+5.7$, and $+1.0$ meV/atom; Sec. III C); these tetramer-only magnitudes are *not* extrapolated to periodic $\mathcal{S}(n)$ because concentration and boundary conditions differ. Alternate-placement repeats will test whether α slopes and $\mathcal{S}_{+3\%}^{\text{tet}}$ signs are placement-sensitive at $\sim 5\%$ nominal doping without direct cross-functional

comparison to the reference column; qualitative claims about opposite N/B α signs remain configuration-specific until that grid converges.

c. Cutoff sensitivity benchmark (Table I). The cutoff400 row isolates the $400 \rightarrow 350$ Ry confound for $6 \times \text{C}_{60}$ N @ $+3\%$ before any $n \geq 6$ design statement is restored.

E. Polaron transport

The Marcus workflow in Figs. 3 and 4 parallels Ref. 4—charged-state geometry optimizations ($E_{\text{opt}}^{\pm}$) plus vertical single-points at the neutral geometry ($E_{\text{vert}}^{\pm}$)—at the same PBE+D3 level as the periodic $\mathcal{S}$ audit (tetramer α in Table IV remains legacy PBE). Recent work shows that charged-supercell and neutral-defect semilocal routes to polaron self-interaction correction are formally equivalent when the polaronic structure is held fixed [24]; our Marcus λ protocol instead follows the charged vertical/adiabatic split of Ref. 4 without piecewise-linear or γ/μ_{DFT} tuning, and remains orthogonal to the $\mathcal{S}$ audit. Where that study linked hopping J and frontier-orbital localization to mobilities in pristine qHP networks, we retain the same observables for comparability (Fig. 4) but do *not* extend their mobility-centric ranking to neutral total-energy margins. Endohedral band-gap tuning [15] and substitutional $\mathcal{S}$ address a second orthogonal knob (cage filling vs. cage substitution) and are not interchangeable for sequential-screen corrections.

F. Limitations and outlook

Three protocol limits are actively benchmarked (Table I; Sec. II F): rigid strain (Table II), tetramer alternate-placement sensitivity (Table III), and mixed 400/350 Ry cutoffs across n (Table I cutoff400 control). All periodic $\mathcal{S}(n)$ use a single fixed dopant-site map on the production supercell; magnitudes are configuration-specific and are not averaged over placement ensembles until alternate periodic maps are computed. Periodic quantitative rankings therefore remain *audit-level* rather than placement-averaged material laws in this revision. The pending alternate tetramer grid (Table III) additionally differs in exchange-correlation treatment from the reference archive (PBE+D3 vs. legacy PBE), so placement and functional confounds are not separated in this revision. Additional limits: tetramer vs. periodic concentration mismatch; PBE+D3 without Koopmans/rVV10; $\mathcal{S}_{\text{gap}}$ not quantified; Mayer bond-order and Bader charge partitioning along the strain path are not computed (Table V reports geometric metrics only); the electron-channel reorganization energy λ^- for B doping is unphysical under the fixed neutral-geometry vertical protocol and is excluded from rate estimates; mobility tensors are not computed—polaron transport anisotropy in related

fullerene networks is treated only comparatively [4, 6] (Figs. 3 and 4).

V. CONCLUSIONS

Sequential strain-doping workflows in covalent molecular networks can omit a meV/atom-scale total-energy cross term when local dopant distortion couples nonlinearly to external biaxial load; under the rigid-strain protocol, quoted $|\mathcal{S}|$ magnitudes serve as upper bounds—not predictions—of equilibrium cross terms until ionic-relaxation and alternate-placement benchmarks complete (Tables II and III; Table II 2/4 fixed-cell corners converged). In qHP C_{60} graphullerene we answered the three Introduction questions under periodic PBE+D3 audits (tetramer α from the legacy PBE archive in Table IV):

1. opposite donor/acceptor α for N vs. B on the archived tetramer grid (Table IV);
2. meV/atom-scale omitted cross terms on a fifteen-point periodic grid ($|\mathcal{S}|$ up to 31.9 meV/atom for P at $n=1$ at fixed coordinates; Table I);
3. $|\mathcal{S}|$ governed by local cage distortion (Table V) rather than tetramer $|\alpha|$ for size-mismatched P.

Ionic-relaxation, alternate-placement, and matched-cutoff controls (Tables I–III) will test whether these rigid-strain upper bounds and configuration-specific magnitudes survive relaxation and placement averaging; on the Table II four-corner grid, converged pristine corners already indicate a much smaller strain increment after fixed-cell ionic relaxation than on the rigid legacy reference (Sec. III C), so $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}|$ is assessed for *sign* retention only until the P -doped corners complete and a matched-functional rigid grid exists. Cou-

pled (ϵ, δ) screening is advisable for size-mismatched substituents and close substitution-energy competitors, complementing strain-only mechanical [13], derivative-network [11], and endohedral band-edge [15] axes. Future work completes the Table II P -doped ionic-relaxation corners and a matched-functional rigid grid for quantitative retention ratios; on Table III, completes the seed-137 alternate-placement PBE+D3 grid and adds PBE+D3 rigid repeats on the reference placement so placement and exchange-correlation confounds are not conflated (reference-placement $\mathcal{S}_{+3\%}^{\text{tet}}$ already tabulated on the legacy archive). Extends placement tests to periodic cells, adds Mayer/Bader strain-path partitioning, and explores polaron-aware semilocal corrections beyond the Marcus vertical protocol.

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DATA AVAILABILITY

The data that support the findings of this study are openly available at <https://github.com/chenxingqiang/graphullerene-strain-engineering>, including converged computational inputs and outputs, tabulated synergy and polaron analyses, and materials to reproduce all figures and tables in the main text. Pending ionic-relaxation, alternate-placement, and matched-cutoff validation tasks are documented in Tables I–III. A tagged release matching the manuscript revision will be issued upon acceptance.

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